

Calvin Joyce

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Objective

Medical robotics R&D / New Product Development internship, **available Jan–Sep 2027**. Graduate researcher under **Prof. Michael C. Yip** (UCSD ARCLab); *Nature* co-author and 5× first author (ICRA, ISRR). Focusing on autonomy, controls, and learning for patient-contact robots, grounded in years of building the hardware myself, deploying on and improving real systems.

Education

University of California, San Diego · B.S. & M.S. Mechanical Engineering (M.S. GPA 3.8/4.0)

B.S. 2024 · M.S. expected June 2027

Relevant coursework: Robotics, Robot Manipulation and Control (in progress), Haptics, Soft Robotics, Linear Control, Mechanical Design

Skills

- **Medical Robotics & pHRI:** haptics & bilateral teleoperation, catheter/continuum robots, clinical safety thresholds, user studies, OptiTrack
- **Software & Control:** Python, C++, MATLAB, PyTorch, ROS 2; imitation learning (ACT), PPO, MPPI; NVIDIA Warp/Newton
- **Hardware:** SolidWorks + FEA, Fusion 360, DFMA, resin/FDM printing, laser cutting; design requirements & acceptance criteria, verification testing, test fixtures, capstan drives, motor drivers, sensors

Research & Work Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego · Advisor: Prof. Michael C. Yip

June 2021 – Present

Grad. Student Researcher (funded, full-time, 2025–); R&D Engineer (full-time, 2024–25); Research Asst. (paid, 2022–24)

La Jolla, CA

• PHABS (Portable Haptic Assisted Bimanual System)

Project lead; first author, ICRA 2027 submission

- * Invented the first handheld bimanual device to render the internal force between two hands on a shared object, the signal that decides crushing or dropping it, by reacting that force across the operator's hands so the device needs no grounding.
- * Closed the force loop with a dual-arm follower, by splitting its contact forces into per-hand pinch and signed inter-hand force, each rendered on its own channel and verified against a reference load cell.
- * Made demonstrations safer, by rendering contact force to the operator: in a 10-person, 247-trial study, pinch feedback cut egg crushes from 33% to 6% and grip force by two-thirds, and the inter-hand channel nearly halved squeeze on a shared-object lift.

• CRIB (Clinical Robotics for the Infant Bedside)

Project lead; first author, ICRA 2027 submission

- * Built the first framework scoring robots against clinical safety limits for infant pickup and CPAP mask repositioning, by measuring safety on the infant rather than the robot, so one pipeline scores humans, teleoperators, and policies alike.
- * Converted clinical practice into testable acceptance criteria with four NICU clinicians: cervical pitch and head acceleration for pickup; per-site contact force, eye/mouth exclusion zones, and a 15 s PEEP budget for CPAP.
- * Benchmarked direct-human, teleoperated, and learned (ACT) handling across 641 instrumented trials on a Unitree G1 and an OpenArm, by building a motion-captured infant manikin and a force-sensing CPAP mask.
- * Showed force regulation is the gap: pickup success ran from 100% (human) to 83% (policy), but on CPAP humans met the pressure bound on just 1.7% of trials, against 61% for expert teleoperation.

• Humanoid Surgery ([video](#)) · *in vivo feasibility of humanoid robots in laparoscopic surgery*

Co-author, *Nature* 2026

- * Enabled humanoid robots to operate standard medical tools in dry-lab and in vivo porcine surgeries, by designing the tool-mount system (optimized mass/CoM, bench- and sim-validated rigidity) and running mechanical setup for every surgical session.
- * Coordinated a 10-person robot-clinician team through live surgeries, turning surgeon feedback into hardware revisions.

• Autonomous Catheter Navigation · *continuum robotics; ARPA-H & DARPA project*

Jul 2026 – Present

- * Building the learned low-level controller that drives a catheter tip to planner-issued waypoints, by training PPO policies in GPU physics on real patient anatomy, with contact limits anchored to measured tissue perforation forces.
- * Found and fixed a physics fault that had silently disabled the catheter's elasticity, by ruling out device, anatomy, drive, and stiffness with data before tracing it to our containment layer.

• Mentorship: 16+ students across lab projects, through CAD/FEA design reviews, DFMA, fixture design, and test plans.

ArtemisAI Labs · Mechanical Engineer, La Jolla, CA

Jan 2026 – Present

- Made a NICU bedside camera mount clinically deployable, by redesigning it against load, retention, and cleanability requirements (IEC 60601-1 instability test, no-disassembly wipe-down); it now fields the cameras for ArtemisAI's Phase II trial at Mount Sinai.

Triton Robotics, UC San Diego · Team President / Advisor

June 2020 – June 2023

- Scaled the team 20→80+ members and diversity 15%→35%+, by formalizing onboarding, DEI partnerships, and targeted recruiting; ran a 17-lead org with OKRs and roadmaps that won national nonprofit status, \$3K in funding, and permanent lab space.

Selected Research Projects

• Haptic Shoulder ([paper](#)) · *pHRI phantom; co-first author, ICRA 2025; provisional patent*

Jan 2024 – June 2024

Invented a biomechanically accurate shoulder phantom for safe pHRI testing, by dedicating one motor-encoder module to each anatomical DOF to reproduce joint center, coupled kinematics, and range of motion.

• ARCSnake V2 ([paper](#)) · *amphibious screw-propelled snake robot, w/ NASA JPL; IROS 2026*

June 2021 – Sep 2025

Increased screw output torque 40% by raising the torque ratio, aligning the belt drive, and cutting friction (shorter drive stack, relieved rubbing surfaces); engineered the mobile testbed that characterized screw performance across media (co-first author, ICRA 2023); designed NASU, the first Archimedes screw with reconfigurable angle of attack (first author, ISRR 2024).

Publications ([Google Scholar](#))

- *Nature* 2026 (co-author): “[In vivo feasibility study of humanoid robots in surgery](#)”
- ICRA 2027 submissions (first author): “PHABS (Portable Haptic Assisted Bimanual System) for Force-Annotated Bimanual Shared Object Teleoperated Data”; “How Safe Is Robotic Infant Handling? Benchmarking Learning from Demonstration Against Clinical Thresholds”
- ICRA 2025 (co-first author): “[Haptic Shoulder for Rendering Biomechanically Accurate Joint Limits for Human-Robot Physical Interactions](#)”
- ISRR 2024 (first author): “[NASU — Novel Actuating Screw Unit: Origami-Inspired Screw-Based Propulsion on Mobile Ground Robots](#)”
- ICRA 2023 (co-first author): “[Mobility Analysis of Screw-Based Locomotion and Propulsion in Various Media](#)”
- Additional: “[Humanoids in Hospitals](#)” (arXiv 2025, co-2nd author); “[LapSurgie](#)” (ICRA 2026); “[ARCSnake V2](#)” (IROS 2026)